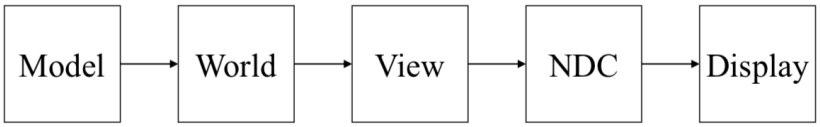
- **Graphics pipeline**, it describes the step-by-step process of converting a model into a displayed image.



  - It consists of five stages:
    1. **Model coordinates** (where object shapes are defined locally)
    2. **World coordinates** (where models are positioned within a global scene)
    3. **Viewing coordinates** (where the scene is projected onto a virtual screen from a camera's perspective)
    4. **Normalized Device Coordinates (NDC)** (where coordinates are standardized between 0 and 1)
    5. **Device Coordinates** (where the image is mapped to the specific resolution of the output device, such as a 1024x768 screen).
- **Polygon** (simple, complex, concave, convex)
  - Polygon is closed polyline where its start point equal to its last point
    - Simple polygon has a well defined two regions(inside and outside)
    - Complex polygon has a self-intersecting sides
    - Concave polygon has at least one diagonal lie outside the polygon
    - Convex polygon hat its all diagonal lies inside the polygon
- **A Raster image is** composed of pixels arranged in a 2D array or bitmap).
    - **Monochrome raster image** (bi-level),
    - **Gay-scale** (where pixel depth determines the number of gray levels, e.g., 8 bits = 256 levels), and
    - C**olor raster images** (where colors are typically represented as combinations of red, green, and blue, with true color using 24 bits per pixel for 16 million colors).
- **Pixel color representation** through **direct coding** and **indexed color with Look-Up Tables (LUT)**.
  - Direct coding stores the full color specification in each pixel (e.g., 24 bits for true color).
  - Indexed color, by contrast, stores only an index into a color palette (LUT), which offers a compromise between low storage requirements and sufficient color support.

Solve the following Problems

- If we use direct coding of RGB value with **5 bits per primary color**, how many possible colors do we have for each pixel?

Solution:

- Levels of red = $2^5 = 32$
- Levels of green = $2^5 = 32$
- Levels of blue = $2^5 = 32$
- Total colors = $32 \times 32 \times 32$ = **32,768 colors**

- If we use a **10-bit pixel value** in a **32-bit lookup table** representation, how many entries does the lookup table have and how much is the size of the lookup table in bytes?

Solution:

- Lookup table entries = 2^{10} = **1,024 entries**
- Lookup table size = $(2^{10} \times 32) / 8 = (1024 \times 32) / 8$ = **4,096 bytes** (or 4 KB)

- A frame buffer supports **12 planes of memory (4 planes for each color)**. Calculate the total number of colors in the palette.

Solution:

- Levels of red = $2^4 = 16$
- Levels of green = $2^4 = 16$
- Levels of blue = $2^4 = 16$
- Total colors = $16 \times 16 \times 16$ = **4,096 colors**

- Compare **System A** (32 bits/pixel frame buffer, no LUT) and **System B** (12 bits/pixel frame buffer with a 30-bit LUT) in terms of memory requirements for a **1920 × 1080** display.

Solution:

System A:

- Frame buffer = $1920 \times 1080 \times 32$ bits
- = 66,355,200 / 8 = 8,294,400 bytes ≈ **8.29 MB**

System B:

- Frame buffer = $1920 \times 1080 \times 12$ bits = 24,883,200 bits = 3,110,400 bytes ≈ **3.11 MB**
- LUT entries = $2^{12} = 4{,}096$
- LUT size = $(2^{12} \times 30) / 8 = (4096 \times 30) / 8$ = **15,360 bytes** ≈ 0.015 MB
- Total System B memory ≈ 3.11 + 0.015 = **3.125 MB**

- A display system uses a **frame buffer with 10 bits per pixel** and a **LUT with 36-bit entries**. The display resolution is **2560 × 1440**.
  a. How many colors can be displayed simultaneously?
  b. How many total colors are available in the palette?
  c. Calculate the total memory required (frame buffer + LUT) in megabytes.
  d. If the system instead used direct coding with **10 bits per primary color**, how much frame buffer memory would be required (no LUT)?

Solution:

a) Simultaneous colors: 10-bit pixel value → 2^{10} = **1,024 colors**

b) Total palette colors: 36-bit LUT → 2^{36} = **68,719,476,736 colors** (≈ 68.7 billion)

c) Total memory:

- Frame buffer = $2560 \times 1440 \times 10$ bits = 36,864,000 bits = 4,608,000 bytes ≈ **4.39 MB**
- LUT size = $(2^{10} \times 36) / 8 = (1024 \times 36) / 8$ = **4,608 bytes** ≈ 0.0044 MB
- **Total ≈ 4.39 + 0.0044 = 4.394 MB**

d) Direct coding with 10 bits per primary color:

- Total bits per pixel = $10 \times 3 = 30$ bits
- Frame buffer = $2560 \times 1440 \times 30$ bits = 110,592,000 bits = 13,824,000 bytes ≈ **13.18 MB**